Phase One Evidence for Bounded Visual Memory

I-JEPA prototype Controlled prediction and equal capacity deletion
Academic evidence report | 16 October 2026

1 Research objective

Our proposal asks which historical observations a model should retain when its memory is limited. The intended learning signal is the prediction cost of deleting an observation: a memory should matter when forgetting it makes useful future prediction worse. The research goal is a better retention decision, rather than a predictor that solves every possible future.

Phase One tests the prerequisites: whether stored visual features contain the missing evidence, and whether a predictor can use it. The later deletion assay adds a bridge toward policy learning, while keeping the predictor and memory capacity fixed. No retention policy has been trained.

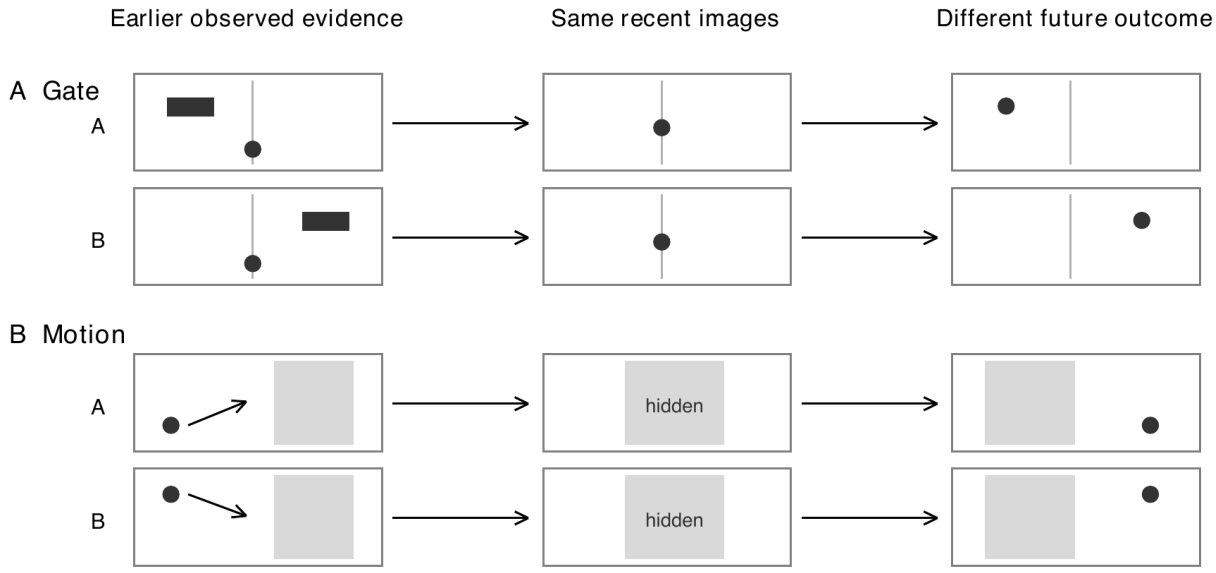


Figure 1. Read each row from left to right: earlier evidence, recent view, then future outcome. A and B look the same recently but have different futures. A useful memory must preserve what distinguishes them: the Gate cue or earlier Motion positions. Drawings illustrate the task; they are not test images or model predictions.

Three questions guide the experiments

Question	Evidence needed
Representation	Can the relevant cue or motion evidence be recovered from frozen image features?
Prediction	Does supplying historical evidence improve forecasts over recent observations alone?
Deletion signal	At the same memory capacity, does losing critical evidence cause more harm than losing a nuisance frame?

Scope. The proposal asks for evidence that history is useful before policy training. It does not require every diagnostic to succeed. We therefore report coverage, effect sizes and exceptions; the original strict all-checks verdict is retained separately. The deletion assay is a later-stage bridge, and no policy has been trained.

2 Experimental design

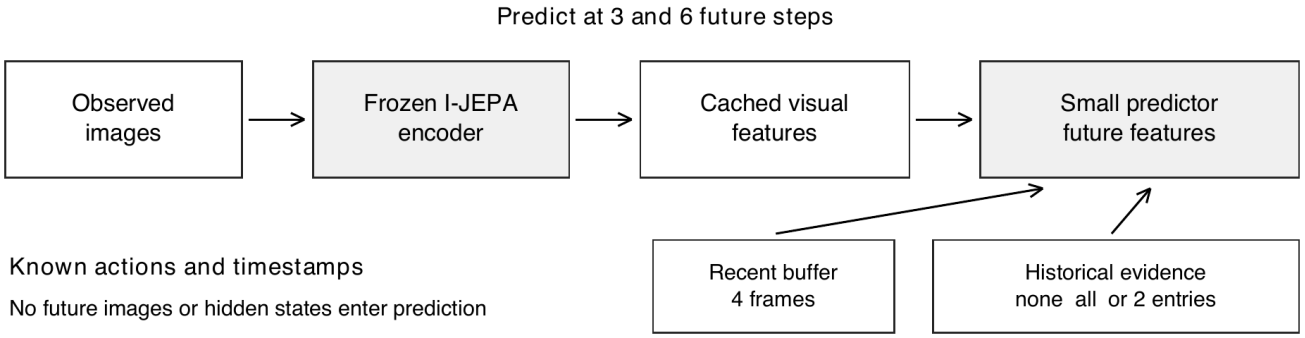


Figure 2. Images become numerical features, then a small model predicts future features. Compare the same recent context with and without history to measure memory value. The image encoder is frozen I-JEPA [1], not V-JEPA. No future image or hidden simulator state is supplied to the predictor.

Component	Specification
Controlled tasks	Gate cue and occluded Motion
Data per task	512 training pairs; 96 validation pairs; 128 test pairs in each of 3 conditions
Test conditions	Ordinary; longer gaps or occlusion; changed visual distractors
Prediction context	4 recent frames; historical budget B = 2; horizons h = 3 and h = 6
Complete candidate models	2 tasks x seeds 17, 29, 43 = 6 models; 120 epochs and 7680 steps each
Visual representation	Frozen I-JEPA; train-only normalization and 128-dimensional basis; full 5120-dimensional scored output
Model selection	Earliest minimum of validation MSE, equally weighting 4 conditions and both horizons
Data separation	Pairs stay in one split; no simulator state, sibling identity or future image is a predictor input

What was compared

Condition	Information provided
Recent only	Recent buffer and actions; no historical evidence
Full history	All available legal history; a diagnostic upper information reference
Privileged small budget	Two important historical frames, chosen using task construction
Content intervention	Replace the relevant historical content to test whether prediction depends on it

The last complete candidate used centered visual differences, attention pooling over observed history and explicit elapsed-time transport. These components were motivated by Set Transformer [2] and NLinear [3]; the candidate is an adaptation, not a reproduction or a novelty claim. Its empty-history output equals the frozen recent-only reference.

Statistics. Feature error is mean squared error (MSE); task error measures lane or position. Reported prediction intervals use paired bootstrap resampling over 128 base trajectory pairs. Seed-averaged intervals are conditional on the three tested training seeds. Forks, siblings and seeds are not treated as extra independent episodes.

3 What the tests actually measure

These labels describe different questions. A pass counts one comparison meeting its fixed effect and uncertainty criteria; it does not mean that an entire model or the project is proven. Some rows overlap. The comparison unit is a condition, horizon and training-run summary, evaluated using 128 matched trajectory pairs.

Label in the summary	Plain question	What we actually do
Evidence readable from features	Did encoding preserve the relevant clue?	Fit a simple linear readout on training features and evaluate held-out features. Decode the Gate cue or Motion direction. This does not test future prediction or precise velocity recovery.
Original prediction checks	Does the model use history to make a better forecast?	Combine feature readouts, two-key-memory versus recent-only prediction, and a wrong-history intervention. Swapping the paired history should move prediction toward the other future.
Additional fixed-reference checks	Does the repaired model improve against a fixed earlier predictor?	Compare its two-memory forecast with the earlier frozen recent-only forecast. Also compare both predictors with the same two memories on the designated hard condition.
Critical deletion contrasts	Does removing useful evidence hurt more than removing a nuisance?	Start from the same three memories and delete each in turn, always retaining two. Compare loss after deleting the Gate cue, or Motion start/end, with loss after deleting nuisance content.
Cross-horizon deletion checks	Is a useful forgetting choice useful at another future time?	Choose the lowest-feature-error deletion at 3 steps and score it at 6; then reverse. Compare with random deletion. This choice uses future outcomes and is not a past-only policy.
Learned-policy advantage and fresh confirmation	Can a deployable scorer beat baselines, and can the result repeat on new data?	Two separate future studies: train a past-only retention scorer and test streaming updates against strong baselines; independently repeat the fixed protocol on untouched data. Neither has been run.

Where the denominators come from

Original 51 checks = 3 feature-readout checks + 24 history-benefit checks + 24 wrong-history checks. Each group of 24 contains 3 test conditions x 2 future times x 4 summaries. The four summaries are three training seeds and their mean, not four independent training runs. The first row of the summary table is therefore already included in this group.

Additional 32 checks = 24 comparisons with the fixed recent-only reference + 8 same-memory comparisons on the hard condition. The latter uses 2 future times x 4 summaries; the hard condition is distractor shift for Gate and long occlusion for Motion.

Deletion contrasts: Gate has 3 conditions x 3 seeds x 2 future times x 1 critical role = 18. Motion has two critical roles, start and end, so it has 36. Cross-horizon checks have 3 conditions x 3 seeds x 2 directions = 18 per task; each requires support in both feature and task error.

A concrete example

For Motion, keep the same recent view and compare two equally sized memories: start plus end, versus end plus shutter. If losing the start makes prediction materially worse, this supports keeping the start. It does not yet show that a scorer can identify that choice without seeing the future, or that its absolute long-occlusion forecast is accurate enough.

4 Prediction gains and coverage

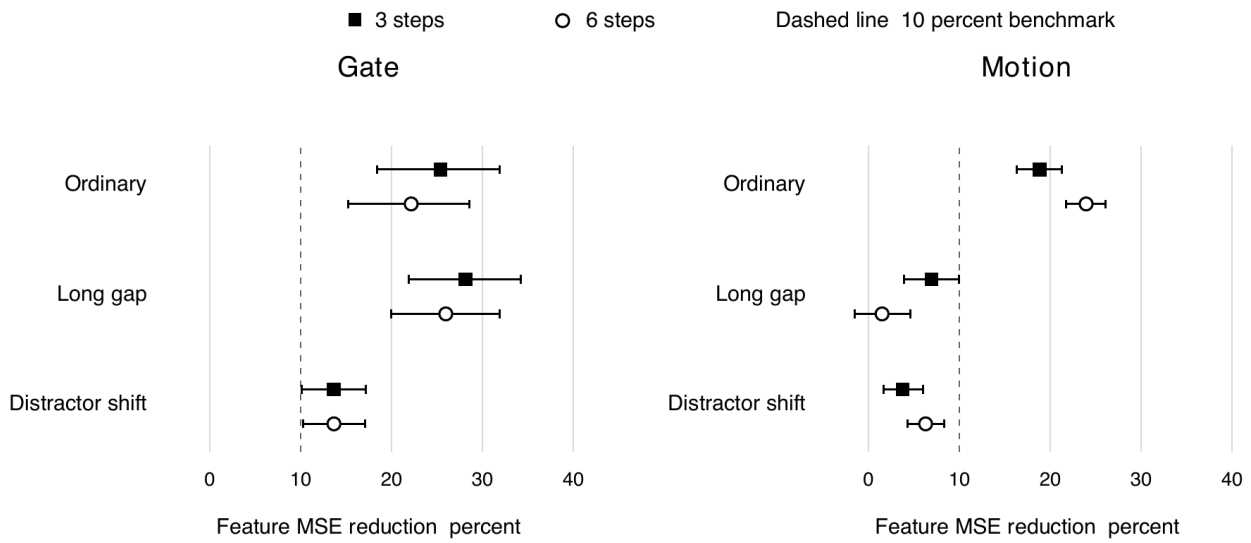


Figure 3. Further right means a larger error reduction from history. A point is the average gain; its horizontal line shows uncertainty. A line crossing zero leaves the direction uncertain. Squares predict 3 steps ahead and circles 6. Gate gains are broad; Motion gains shrink in harder conditions. Paired 95 percent intervals, conditional on the three seeds; D1.

Task and condition	Recent only MSE	Full history MSE	Budget 2 MSE	Gain and 95 percent CI
Gate Ordinary	0.0997	0.0562	0.0776	22.2% [15.2, 28.5]
Gate Long gap	0.0995	0.0960	0.0737	26.0% [20.0, 31.9]
Gate Distractor shift	0.1385	0.1003	0.1196	13.7% [10.3, 17.1]
Motion Ordinary	0.1526	0.0995	0.1161	23.9% [21.7, 26.1]
Motion Long gap	0.1831	0.1879	0.1804	1.5% [-1.5, 4.6]
Motion Distractor shift	0.2908	0.2713	0.2726	6.3% [4.3, 8.3]

Six-step normalized feature errors, averaged over three seeds. The last column compares privileged two-memory prediction with recent only. Full history uses more memory and is an information reference, not a capacity-matched competitor. Data: D1.

Evidence check	Gate coverage	Motion coverage
Evidence readable from features	3 / 3 pass	3 / 3 pass
Privileged history benefit under original task rules	23 / 24 pass	8 / 24 pass
Response to historical content intervention	24 / 24 pass	24 / 24 pass
Fixed recent-reference feature comparisons	24 / 24 pass	8 / 24 pass
Critical error improvement over prior predictor at the same budget	8 / 8 pass	0 / 8 pass

Table 1. x/y pass means x of y comparisons met the fixed criterion. Counts overlap in data and measure different claims, so they are not a probability of success or a project completion score. D1.

Interpretation. Gate has broad coverage: 50/51 original and 32/32 additional checks, with one inconclusive ordinary h6 result for seed 43. Motion has 35/51 and 8/32, with limitations concentrated in long occlusion and changed distractors. These are useful, uneven findings. The earlier strict all-checks rule was not met; that record is unchanged.

5 Does deletion cost identify important memory

We reused the six complete models and their existing feature caches. From the same parent set of three real observations, we removed each entry in turn and compared the resulting forecasts. Every fork retained exactly two historical entries, so a gain cannot be explained by a larger memory budget.

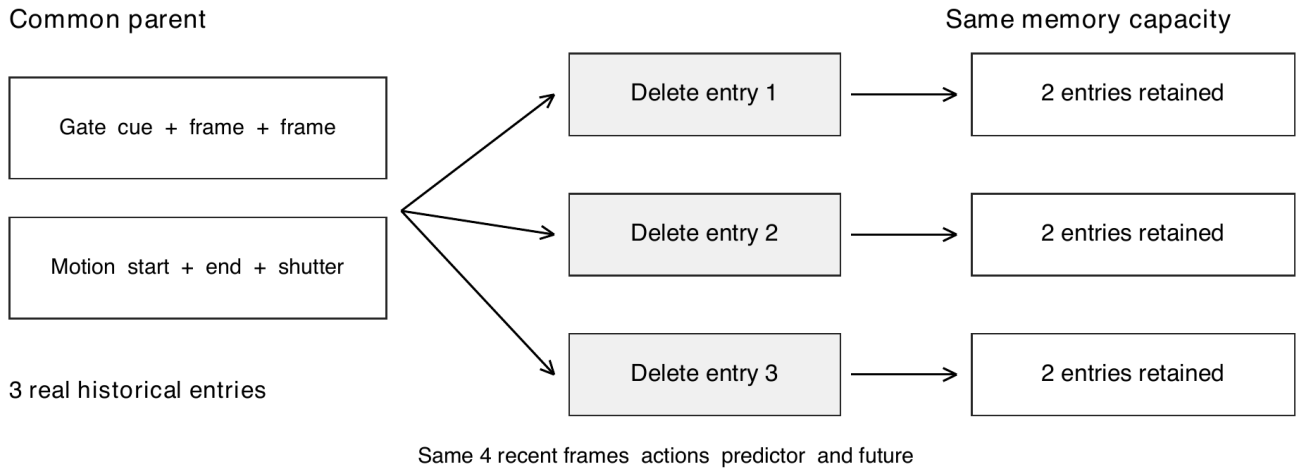


Figure 4. Start with the same three memories, remove a different one in each branch, and always leave two. If removing the important observation hurts more, deletion cost has identified useful evidence rather than extra capacity. Two fixed parent variants were used. These constructed pools are diagnostics, not a learned policy.

A check is counted when keeping critical evidence reduces both visual-feature error and lane or position error by at least 10 percent, with a positive paired 95 percent interval. Each check uses 128 base pairs. Gate has one critical role and Motion two, giving 6 and 12 checks per condition. The thresholds and original results remain unchanged.

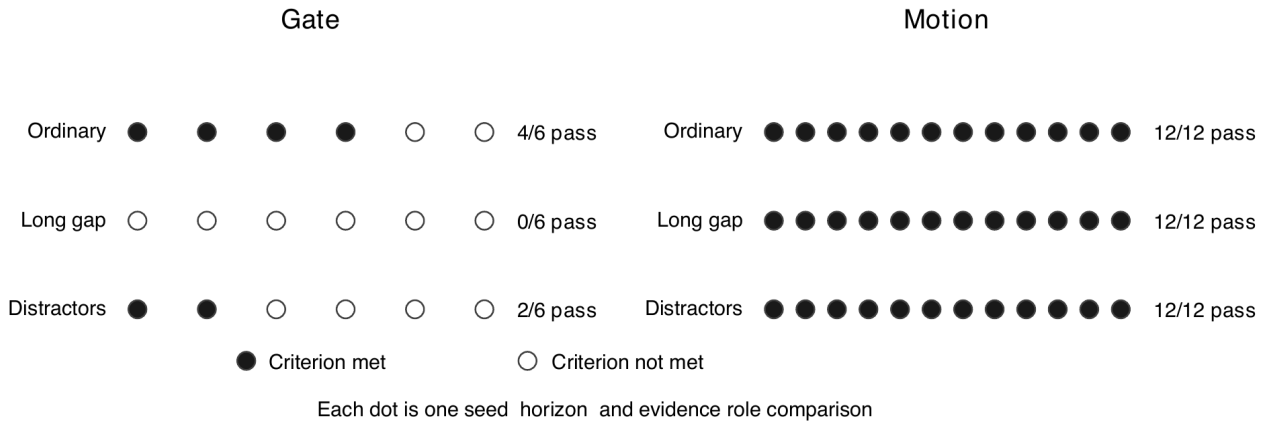


Figure 5. Filled dots count comparisons meeting the criterion; open dots count those that did not. Dot order has no meaning. Motion has 36/36 across conditions. Gate has 4/6 in ordinary conditions, 0/6 with long gaps and 2/6 with distractors: partial support with clear limits, not a single binary outcome. D2.

Task	Ordinary	Long gap	Distractor shift	Cross horizon checks
Gate	4 / 6 pass	0 / 6 pass	2 / 6 pass	18 / 18 pass
Motion	12 / 12 pass	12 / 12 pass	12 / 12 pass	18 / 18 pass

Table 2. Cross-horizon checks ask whether a deletion chosen at 3 steps is useful at 6, and vice versa, relative to uniform deletion. Both tasks show 18/18 coverage here, while Gate has weaker critical-versus-nuisance coverage. Selection uses future outcomes, so this is not yet an online policy result. D2.

6 Effect sizes and uncertainty

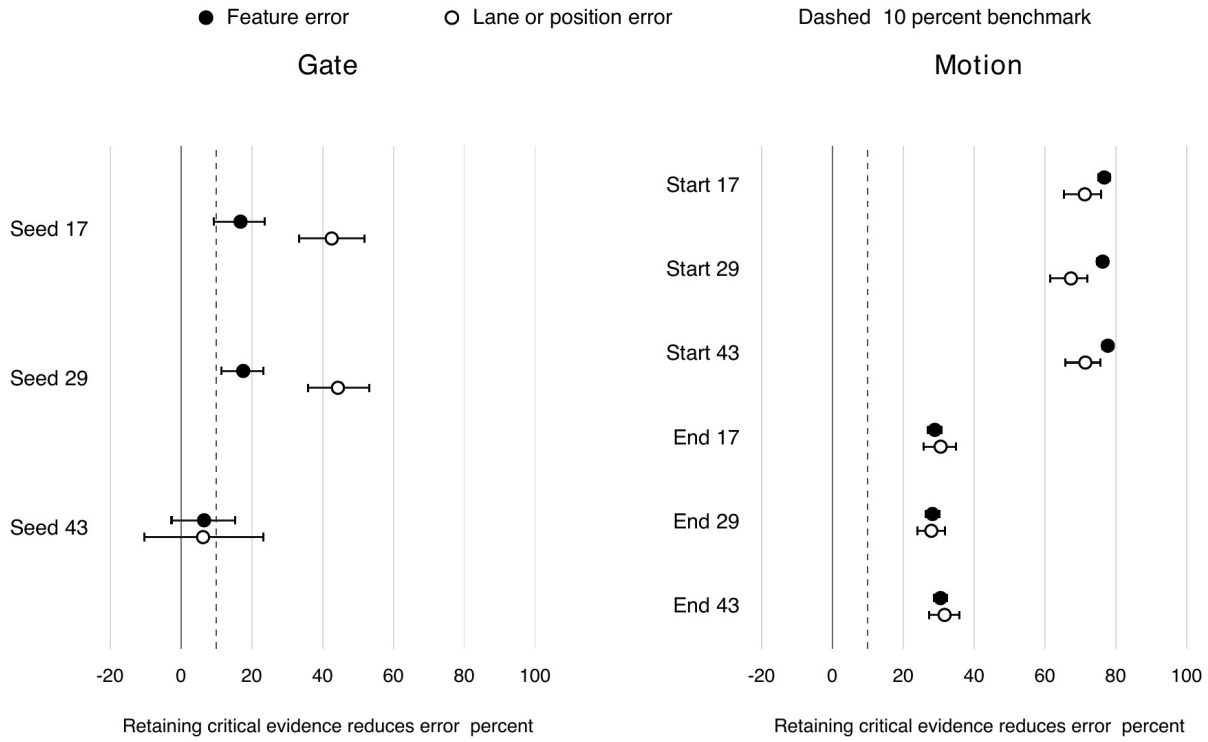


Figure 6. Further right means keeping that memory is more useful. Black dots measure feature error; white dots measure lane or position error. Lines show 95 percent uncertainty. Motion start and end memories help in all three runs. Gate seed 43 lies near zero, showing uncertainty rather than evidence of no value. Ordinary condition, 6 steps ahead; D2.

Selected contrast	Delete critical feature MSE	Delete nuisance feature MSE	Difference and 95 percent CI
Gate cue seed 43	0.0987	0.0923	0.0065 [-0.0026, 0.0152]
Motion start seed 17	0.5005	0.1163	0.3842 [0.3529, 0.4162]
Motion end seed 17	0.1636	0.1163	0.0473 [0.0430, 0.0519]

Table 3. Illustrative numeric contrasts at h6. Within each row, only the deletion differs. Losses across Gate and Motion are not calibrated to the same task difficulty. All seeds and both Motion roles are shown in Figure 6.

Motion. Coverage is 36/36 critical comparisons and 18/18 cross-horizon checks. At h6, retaining the start observation reduced feature error by 76.3 to 77.8 percent, and retaining the end observation by 28.3 to 30.5 percent. Position-error contrasts agreed. Both observations carry useful complementary evidence.

Gate. Ordinary coverage is 4/6. The two exceptions are seed 43: at h3 lane benefit is uncertain; at h6 feature benefit is 6.5 percent and lane benefit 6.3 percent, with both intervals crossing zero. This limits claims of seed stability without erasing the four supported comparisons.

Different questions. Deletion cost asks which forgetting choice is worse. Prediction quality asks how accurate the forecast is overall. Motion can provide a consistent ranking of memory value while still forecasting difficult long occlusions poorly. These results describe different capabilities.

7 A useful signal does not prove a better policy

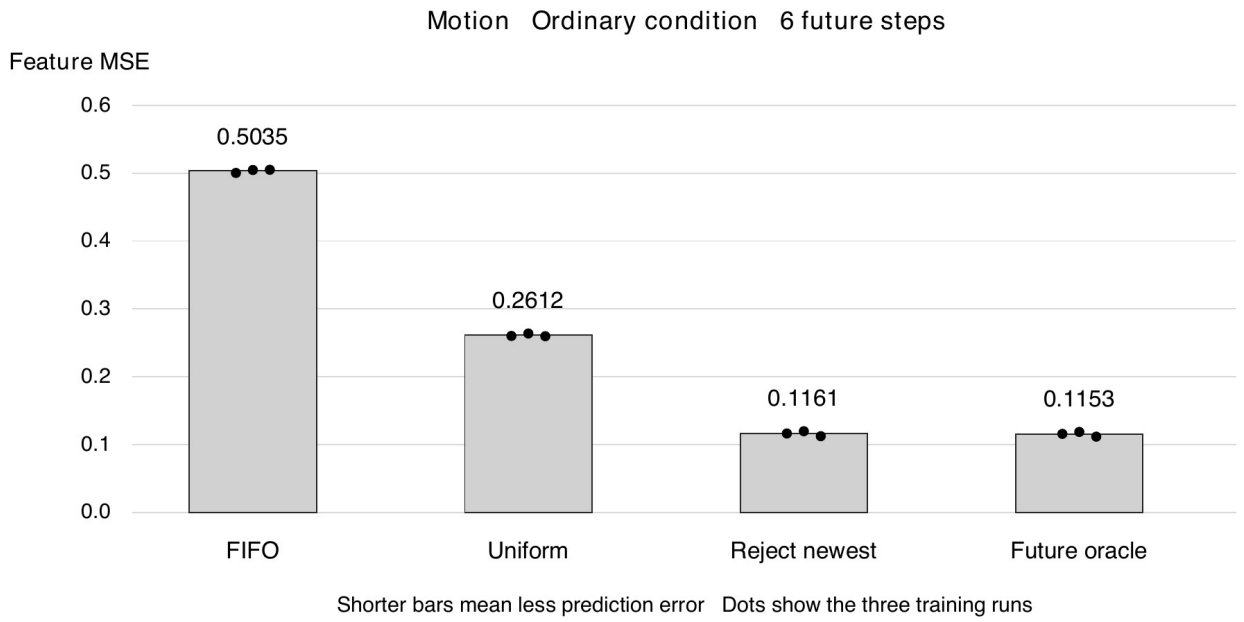


Figure 7. Shorter bars are better. FIFO removes the oldest entry; uniform deletion chooses randomly; reject-newest keeps the existing memories. The future oracle uses unavailable future outcomes to choose the best removal. Reject-newest nearly matches this ideal reference in these simple pools. Bars average three runs; dots show each run. D2.

Deletion rule	Feature MSE	Position MSE
FIFO	0.503474	0.025186
Uniform	0.261175	0.014506
Reject newest	0.116061	0.007548
Future oracle	0.115305	0.007338

The oracle feature MSE is only 0.65 percent lower than rejecting the newest frame: 0.115305 versus 0.116061. This small headroom matters. These constructed pools show which evidence is useful, but a simple rule already finds most of that value; they do not yet demonstrate a need for a learned selector.

What the experiments have achieved

We have a working visual-feature prediction pipeline, paired causal input comparisons, auditable model checkpoints, and a fast same-capacity deletion evaluation. Motion supplies a reproducible signal about relative memory value. Gate supplies a concrete counterexample to assuming that feature-level utility is always stable across training seeds.

Limits that should shape the next experiment

Adaptive development data. The same development trajectories informed predictor repairs. No untouched-data confirmation has been run.

Privileged pools. Candidate times were constructed using known task evidence. This does not show that a policy can discover them from past inputs alone.

Single deletion. Every fork resets to its parent. Continuous irreversible forgetting, accumulated errors and long-term redundancy remain untested.

Uncertain learning headroom. A learned policy must beat strong past-only rules in more varied memory states, not merely beat random deletion in an easy diagnostic pool.

8 Evidence coverage and next research scope

Evidence category	Gate	Motion
Evidence readable from features	3/3 pass	3/3 pass
Original prediction checks including the feature checks above	50/51 pass Broad coverage	35/51 pass Limited on harder tests
Additional fixed-reference checks	32/32 pass	8/32 pass
Critical deletion contrasts across all conditions	6/18 pass Ordinary 4/6	36/36 pass Ordinary 12/12
Cross-horizon deletion checks	18/18 pass	18/18 pass
Learned-policy advantage and fresh confirmation	Not yet tested	Not yet tested

Why majority coverage is useful but not sufficient by itself

Exploratory research does not require universal success. High coverage supports a broad but qualified claim; concentrated exceptions identify its boundary. However, correct pairing, past-only inputs and equal capacity are validity requirements, not votes. A large count of easier successes cannot establish long-occlusion reliability when that specific condition is weak. These checks also share data, so 50/51 is not a 98 percent probability that the proposal works.

The earlier all-checks acceptance rule was conservative and remains unmet. This descriptive coverage summary was written after observing the results; it does not replace that rule with a new majority threshold or claim a retrospective confirmation pass.

Evidence and reproducibility

D1 contains six complete models, 720 epoch records and 69120 evaluation rows; 64 fixed comparisons were independently checked. D2 contains 55296 fork-horizon rows and 180 independently recomputed comparisons. The statistical units are 128 base pairs per task and condition, not the raw rows. D2 used zero new GPU time. Settled GPU use was 3 h 32 min 03 s within the 8 h cap; no retention policy or fresh confirmation was run.

Sources

[1] Assran et al. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. CVPR, 2023. https://arxiv.org/abs/2301.08243

[2] Lee et al. Set Transformer A Framework for Attention-based Permutation-Invariant Neural Networks. ICML, 2019. https://proceedings.mlr.press/v97/lee19d.html

[3] Zeng et al. Are Transformers Effective for Time Series Forecasting. AAAI, 2023. https://arxiv.org/abs/2205.13504

D1 Predictor experiment 7574 and audited evaluation 7575/7576, archived in phase1_predictor_repair_v2. This report covers the last complete candidate, not incomplete later trials.

D2 Deletion assay 7590, archived in phase1_deletion_assay. Its protocol preceded deletion outcomes, but its models and data were already development-selected.

Conclusion

The proposal has meaningful, graded support. Gate shows broad prediction evidence at 50/51 plus 32/32 checks, with an unresolved seed and weaker deletion coverage. Motion shows consistent deletion value at 36/36 plus 18/18 checks, while its harder prediction conditions remain limited. This supports a scoped Motion learnability pilot and targeted Gate stability checks. It does not yet establish a learned retention advantage, V-JEPA transfer or untouched-data confirmation. The useful outcome is a clearer mechanism and research boundary, not a single pass or fail label.